

DIGITAL FORENSICS EXAMINATION REPORT

BCSSL Lab 14 | Case: CASE-TEST-01 | Strict Chain of Custody Verified

Case Number:	CASE-TEST-01	Examiner:	Digital Forensics Examiner
Evidence File:	space_usb.img	File Size:	128.00 MB
Intake UTC Time:	2026-09-30 13:39:23 UTC	Engine Mode:	The Sleuth Kit (TSK) Native Pipeline
Acquisition SHA-256:	d18852de7077533e0f7dba8743c078cc748c4797d1165706d3d4b065b85ee8ed	Integrity Result:	PASS

CRYPTOGRAPHIC EVIDENCE INTEGRITY AUDIT: PASS

CRYPTOGRAPHIC INTEGRITY VERIFICATION SUCCEEDED: Neither the original evidence nor the working copy was altered during examination. Forensic read-only protection (software attribute; a hardware or driver-level write blocker would be used on real evidence) and custody principles adhered to strictly.

Original: d18852de7077533e0f7dba8743c078cc748c4797d1165706d3d4b065b85ee8ed

Working: d18852de7077533e0f7dba8743c078cc748c4797d1165706d3d4b065b85ee8ed

1. Partition Layout & Filesystem Architecture

Slot / Offset	Start Sector	Length	Description	File System
000	8192	196608	ASTRA9_USER	FAT32 (N/A)
001	204800	49152	ASTRA9_CACHE	Ext2 (.cache)

2. Deleted File Analysis & Forensic Recovery

Deleted Inodes Found: 42 | Recovered Files: 47 | Extension Mismatches (Disguised Files): 1

File Name	Size	Detected Real Type	Status / Mismatch Flag
deleted_inode_17_OrphanFile-17	32.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_18_OrphanFile-18	9.00 B	Plain Text / Document / Script	Intact recovered file
deleted_inode_19_OrphanFile-19	663.00 B	Plain Text / Document / Script	Intact recovered file
deleted_inode_20_OrphanFile-20	16.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_21_OrphanFile-21	66.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_22_OrphanFile-22	84.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_23_OrphanFile-23	70.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_24_OrphanFile-24	69.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_25_OrphanFile-25	77.00 B	Unknown Binary Stream / Fragme	Intact recovered file

deleted_inode_26_OrphanFile-26	65.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_27_OrphanFile-27	86.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_28_OrphanFile-28	82.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_29_OrphanFile-29	57.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_30_OrphanFile-30	78.00 B	Unknown Binary Stream / Fragme	Intact recovered file
deleted_inode_31_OrphanFile-31	74.00 B	Unknown Binary Stream / Fragme	Intact recovered file

... and 32 additional recovered files listed in full HTML report and CSV package.

3. Keyword Content Search Results

Keywords Searched: password | **Total Hits:** 0

No keyword occurrences found matching the specified terms.

4. Forensic Limitations & Anti-Forensics Analysis

Normal Forensic Limitations:

- Inode records where clusters were zeroed or reassigned cannot be fully reconstructed; this is a known physical constraint of post-deletion state rather than a recovery fault.
- MACB timestamps reflect file system metadata stored prior to imaging; FAT timestamps have 2-second resolution for modified times and lack creation seconds.

Anti-Forensics & Deception Detection:

- File extension mismatch analysis identified files where user-facing extensions were intentionally altered to mask underlying document or binary payloads.
- All recovered files are untrusted; execute bits are disabled to mitigate weaponized payload hazards.